

First Terminal Examination - 2079

Class : XII

Subject: Physics (1021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

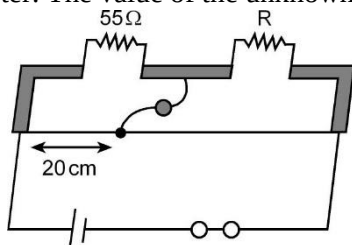
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. $[11 \times 1 = 11]$

1. Figure below shows a meter bridge set up with null deflection in the galvanometer. The value of the unknown resistance R is:



- | | |
|----------------|------------------|
| a. 55Ω | b. 13.75Ω |
| c. 110Ω | d. 220Ω |
2. If the cold junction of a thermocouple is θ_0 and hot junction is kept at $\theta^\circ\text{C}$, then the relation between neutral temperature (θ_n) and temperature of inversion θ_i is
- | | |
|--------------------------------------|--------------------------------------|
| a. $\theta_n = \theta_i$ | b. $\theta_n = 2\theta_i + \theta_0$ |
| c. $2\theta_n = \theta_i - \theta_0$ | d. $2\theta_n = \theta_i + \theta_0$ |
3. In a Wheatstone's bridge, if the battery and galvanometer are interchanged, the condition for balance
- is disturbed
 - is not disturbed
 - depends upon value of resistances of bridge
 - depends upon resistance of galvanometer

4. A potentiometer wire is 10 m long. It has a resistance of $20\ \Omega$. It is connected in series with a battery of emf 3V and negligible internal resistance. The potential gradient along the wire in Volt/m is
- 0.003
 - 0.3
 - 3.0
 - 1.3
5. A monoatomic gas of n -moles is heated from temperature T_1 to T_2 under two conditions (i) at constant volume and (ii) at constant pressure. The change in internal energy of the gas is
- more for (i)
 - more for (ii)
 - same in both cases
 - independent of number of moles
6. The potential difference across a conductor is doubled, the rate of generation of heat will be
- one fourth
 - halved
 - doubled
 - four times
7. For a simple pendulum graph between ' T^2 ' and ' l ' comes out to be
- parabola
 - straight line
 - circle
 - ellipse
8. A moving coil galvanometer has a resistance of 90Ω . If only 10% of the main current may flow through the galvanometer, in what way and which resistance is to be used?
- $10\ \Omega$ in series
 - $810\ \Omega$ in series
 - $10\ \Omega$ in parallel
 - $810\ \Omega$ in parallel
9. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively successively illuminate a metal of work function 0.5 eV. The ratio of maximum speeds of the emitted electrons will be
- 1: 5
 - 1: 4
 - 1: 2
 - 1:1
10. What is the formula of radius of gyration?
- $K^2 = \frac{I}{M}$
 - $K^2 = \frac{I^2}{M}$
 - $K = \frac{I}{M}$
 - $K = IM$

11. The heating element of an electric heater should be made with a material which should be
- high specific resistance and high melting point
 - high specific resistance and low melting point
 - low specific resistance and low melting point
 - low specific resistance and high melting point

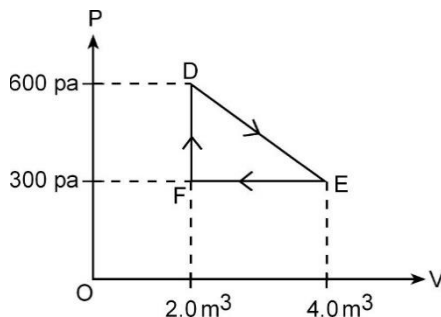
Group - B

Give short answer to the following questions.

[8 × 5 = 40]

12. a. What is the principle of potentiometer? Explain, how can you measure the internal resistance of a cell by using potentiometer? [3]
- b. The emf of a battery A is balanced by a length of 75 cm on a potentiometer wire. The emf of a standard cell, 1.02 V, is balanced by a length of 50 cm. What is the emf of A? [2]
13. The equation $F = -kx$, where k is a constant, is an expression for a law that governs the motion of a body.
- Give the name for this type of motion and describe the motion. [1]
 - When a sphere of mass 500 g is attached to a spring of length 300 mm, the length of the spring increases to 330 mm. Calculate the spring constant. [2]
 - Calculate the time period of the system. [1]
 - Write the positions at which the velocity and acceleration become maximum. [1]

14. a. A thermodynamic system is taken from initial state D to intermediate state E by the linear process as shown in figure. If volume is then reduced to the original value from E to F by an isobaric process.

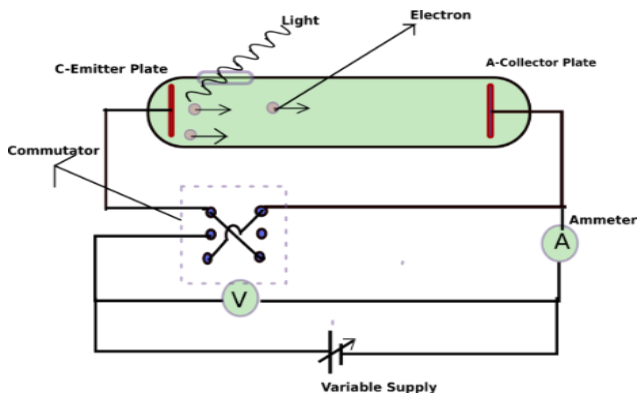


Calculate

- i. the workdone by gas in expanding from D to E. [2]
- ii. total workdone in one cycle. [2]

- b. The workdone from F to D is zero. Why? [1]

15. Study the following diagram and answer the following questions.



- a. What happens to the number of emitted photoelectrons if intensity of light increases? [1]
 - b. What happens to the K.E. of emitted photoelectrons if intensity of light increases? [1]
 - c. What happens to the kinetic energy of emitted photoelectrons if wavelength of incident photon increases? [1]
 - d. What happens to the photoelectric current if the negative potential of collector plate gradually increases and why? [2]
16. a. Define radius of gyration. [1]
- b. Why are spokes fitted in the bicycle? [1]

- c. Define moment of inertia and establish the relation between external torque applied and moment of inertia of body ($\tau = I\alpha$). [3]

17. a. An electron is projected at right angle to the direction of electric field.
- Show that the nature of path followed by electron is parabolic. [2]
 - What will be the nature of path if electron is projected in direction parallel to electric field? [1]
- b. What is cross field? Compare the specific charge of electron and proton. [2]

18. a. Fig. (1) shows the variation of displacement x with time ' t ' in a progressive wave. [4]

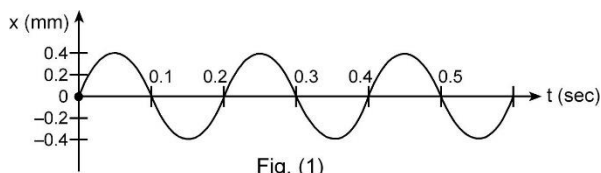
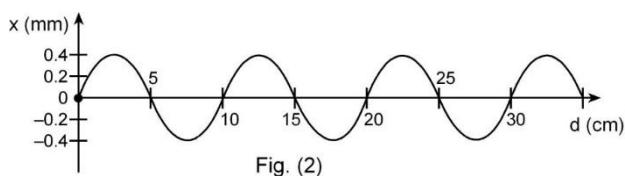


Fig. (2) shows the variation with distance d along the same wave of the displacement x .



Use figures (1) and (2) to determine

- the amplitude
 - the wavelength
 - the frequency
 - the speed
- b. Explain why frequency is the most fundamental property of a wave. [1]

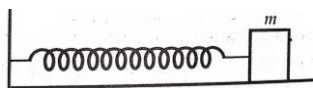
19. a. What is a Superconductor? Write its any two applications. [2]
- b. A Galvanometer coil has resistance 50Ω and a resistor of $1.15K\Omega$ is connected in series. It can read potential difference upto 12 Volts. If the same coil is used to construct an ammeter which can measure currents upto 2A, what should be the resistances of the shunt used? [3]

Group - C

Give long answer to the following questions.

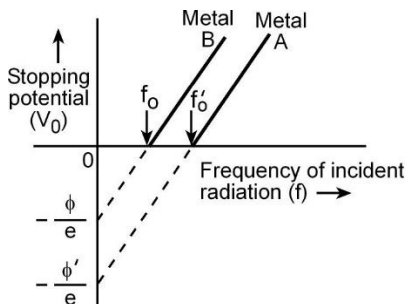
[3 × 8 = 24]

- 20 a. A simple pendulum is given a small displacement from its equilibrium position and performs simple harmonic motion.
- i. State what is meant by simple harmonic motion. [1]
- ii. Define simple pendulum and obtain expression for its time period. [3]
- b. A spring of force constant $1200N/m$ is mounted horizontally on a horizontal table. A mass of $3kg$ is attached to the free end of spring and pulled sideways to a distance of $2cm$ and released. Find



- i. frequency of oscillation of mass [1]
- ii. maximum acceleration of mass [1]
- iii. maximum speed of mass [1]
- c. Define free oscillation and damped oscillation. [1]

21. a. An electron is a negatively charged subatomic particle having charge $1.6 \times 10^{-19} \text{C}$. Describe an experiment to determine the charge of an electron. [3]
- b. The graphs shows the variation of stopping potential (V_0) with frequency of incident radiation (f) for two photosensitive materials A and B having threshold frequencies $f'_0 > f_0$ respectively.



- Which of the two metals, A or B, has higher work function? [1]
 - What information do you get from slope of the graphs? [1]
 - What does the value of the intercepts of graph 'A' on the potential axis represent? [1]
- c. 400nm wavelength of light falls on photosensitive material of work function 2.3eV. Compute the maximum energy of photoelectrons. [2]
(Given Planck's constant = $6.62 \times 10^{-34} \text{ Js}$,
Electronic charge = $1.6 \times 10^{-19} \text{ C}$)
22. a. Why is Sb-Bi thermocouple preferred to Fe-Cu thermocouple? Discuss the variation of thermoelectric emf in a thermocouple with the increase in temperature of hot junction. [1+3]

b. In the adjacent circuit find:

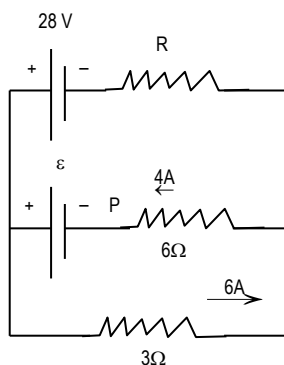
[4]

i. the current in resistor R.

ii. resistance R,

iii. the unknown emf ε ,

iv. if the circuit is broken at P, what is the current in resistor R?



∞ - The End - ∞